AI Product-Image Generation — Current Architecture & Display Audit

Audit date: 24 Oct 2026 · Scope: read-only (code, production DB `SELECT`s inside `BEGIN READ ONLY`, S3 `HEAD` requests, log reads). Nothing was changed.

Evidence labels used throughout: **Confirmed** (seen in code *and* data/logs), **Highly likely** (code path proven, user impact inferred), **Possible** (plausible, not proven).

1. Project information

Item	Value (verified from code/config)
Project path	`C:\Users\hello\OneDrive\Desktop\projects\box-gold`
Repository	`tazio-creatives/BOX-Gold-` (origin); npm workspaces monorepo: `backend/`, `web/`, `admin/`, `packages/`
Branch / latest commit	`main` @ `7bf89ec` — "Product cards: no badge for zero stock (made to order)" (20 Oct 2026)
Working tree	Clean at audit start (this document is the only new file)
Backend	Node 22, Express (ESM), `backend/src` — controllers / services / repositories / jobs / providers
Storefront	React 19 + Vite 8, SSR via Express (`web/server/index.js`), TanStack Query
Admin	React 19 + Vite SPA (`admin/src`), TanStack Query, served as static files by nginx
Database	PostgreSQL 18.6, migrations via node-pg-migrate (`backend/src/db/migrations`)
Queue / worker	**pg-boss v10** (Postgres-backed), runs *inside* each backend process (`jobs/queue.js`); pm2 cluster ×3 → 3 competing workers
Image storage	AWS S3 bucket `boxgold` (ap-south-1), public-read, virtual-hosted URLs (`STORAGE_PROVIDER=s3`, no `S3_PUBLIC_BASE_URL`)
CDN / caching	Cloudflare in front of the site and API (`cf-cache-status: DYNAMIC` — HTML/API not edge-cached). Images are served **directly from S3**, not through Cloudflare. SSR HTML cached in Postgres table `page_cache` (TTL `SSR_CACHE_TTL_SECONDS=600`)
AI image provider	OpenAI: `gpt-image-2` (`images.edit` / `images.generate`, 1024×1024, quality `medium`, PNG); vision/analysis/validation `gpt-4o` (structured output via zod)
Legacy AI pipeline	`ai-images` routes + `aiImageJob.js` with `AI_IMAGE_PROVIDER=stub` — still mounted, not a real generator

Secrets (OpenAI key, DB URL, AWS keys) were present but not read or printed.

2. Current architecture (summary)

There are **three** image-producing paths that all end in `product_images`:

1. **AI Image Studio** (the real one; this audit's focus) — `aiStudio.controller.js` → `aiStudioJob.js` (pg-boss) → `aiStudioService.js` (OpenAI) → S3 `products/ai-studio/<jobId>/…png` → admin import → `imageProcessingService.processAndStoreImage` → S3 `products/<productId>/<uuid>-<variant>.<ext>` → `product_images`.
2. **Legacy "AI images"** (`aiImages.controller.js`, `aiImageJob.js`, `ai_generated_images`, provider = `stub`) — still routable, generates nothing real.
3. **Manual upload / attach existing / Product Size image** (`productImages.controller.js`, `productSizeImageJob.js`).

All three write `product_images` with different "who becomes primary" rules (see §12, H-5).

3. Architecture diagram



Failure points marked in the diagram:

Code	Where	What can happen
L1	analyse	Job fails ("busy") when OpenAI credits are exhausted or rate-limited
L2	`images.edit`	Asset FAILED: credits/429, safety 400, 5xx; cancelled job leaves assets `GENERATING` forever
L3	worker restart/expiry	Assets left `GENERATING` / `PENDING`; job never reaches a terminal status (no reaper)
V1	validation	Rejected by validation; false failures auto-deselect the image
N1	import selection	**Generated but not imported** — deselected assets are silently skipped
K1	storage key	Key is random UUID under job folder (no slot in key — slot lives only in DB)
O1	import order	Gallery position decided by browser import order; retries append out of order
O2	PDP	PDP opens on lowest `sort_order`, not on the featured/primary image
P1	primary	Primary only moves if the featured asset itself is imported; listing image ≠ PDP first image
C1	cache	No cache invalidation on any image mutation → stale SSR HTML up to 10 min

Not a failure point (verified): S3 object existence, MIME types, URL format (§9).

4. Complete generation sequence

#	Step	File · function	Endpoint / job	Tables	Input → Output	Errors / retry
1	Admin opens studio	`admin/src/pages/products/AiImageStudioPage.tsx`	`GET …/ai-studio/active` (resume)	`ai_studio_jobs`	productId → jobId|null	none
2	Upload reference (1 primary + ≤3 supporting)	`UploadStep.tsx` → `api/aiStudio.ts` `createStudioJob` → `aiStudio.controller.js` `createJob`	`POST /admin/products/:id/ai-studio` (multer `uploadEnhanceImage.array('images',4)`, `aiStudioRateLimiter` 20/15 min)	`ai_studio_jobs` (status `analysing`)	files → `reference_image_urls[]`	sharp validation (jpeg/png/webp, 200–8000 px); 409 if an active job exists (partial unique index `ai_studio_jobs_one_active_per_product`)
3	Reference stored	`createJob`	—	—	re-encoded JPEG q92, `products/ai-studio/<productId>/references/<uuid>.jpeg`	S3 error → 500
4	Reference analysis	`aiStudioJob.js` `analyseHandler` → `aiStudioService.analyseJewellery`	pg-boss `ai-studio-analyse`	`ai_studio_jobs.analysis`, `ai_detected_category`, `ai_detected_customer_category`	**primary image only** → `AnalysisSchema`	any error → job `failed` with mapped message; no retry (pg-boss default retry only on throw — handler catches everything)
5	Category detect/correct	`AnalyseConfirmStep.tsx`, `generationRules.inferJewelleryTypeFromCategory`	client state	—	product category name (keyword match) **wins over** AI type → `jewelleryType`	requires explicit confirm click
6	Customer category	`AnalyseConfirmStep.tsx`	client state	—	AI suggestion prefilled → `customerCategory`	explicit confirm required
7	Presenter	`PresenterStep.tsx` (non-Ring only)	`GET /admin/ai-studio/presenters`	`presenters` (2 rows, both active)	→ `presenterId` or null	Ring: forced null server-side
8	Rose Gold	`AiImageStudioPage` (`localStorage` `aiStudio.defaultGenerateRoseGold`, default **on**); `effectiveGenerateRoseGold` (Necklace forced on)	client → confirm body	`ai_studio_jobs.generate_rose_gold`	boolean	—
9	Output manifest	`aiStudioService.resolveAssetTypesForJob` (+ duplicate in `admin/…/generationRules.ts`)	inside `confirmJob`	`ai_studio_assets` rows (`UNIQUE(job_id, asset_type)`, `display_order` CHECK)	flags → asset type list	—
10	Prompt assembly	`buildAssetPromptSections` + `assemblePrompt`; template from `ai_studio_category_templates`	`POST …/prompt-preview` (preview), persisted in `confirmJob`	`ai_studio_assets.assembled_final_prompt`, `custom_creative_instructions`, `prompt_mode`	→ one string per asset	—
11	Submit	`confirmJob` → `boss.send(ai-studio-generate)`	`POST …/confirm`	job `generating`	—	409 if not `awaiting_confirmation`
12	Generate	`aiStudioJob.generateHandler` → `generateOneAsset` → `aiStudioService.generateShot`	pg-boss `ai-studio-generate` (expire 30 min)	asset `GENERATING`	ref JPEG [+ presenter photo | + Gents identity PNG] → PNG 1024²	OpenAI SDK `maxRetries:1`, timeout 600 s; errors → asset `FAILED`, `retry_count+1`
13	Validate	`validateGeneratedImage` (gpt-4o)	same job	`validation_status`, `validation_result`	generated + reference (+identity) → passed/warning/failed	validator error swallowed → status `null`; Ring hand shots auto-retry once on `failed`
14	Store raw	`storageProvider.save`	same job	asset `READY`, `image_key`	PNG → `products/ai-studio/<jobId>/<uuid>.png`	S3 error → asset FAILED
15	Default featured	`ensureDefaultFeatured`	end of run	`is_featured`	first READY `YELLOW_FRONT` / `RING_GOLD_FRONT`	none if that asset failed
16	Job status	`deriveJobStatus`	end of run	job `review_ready` / `partially_failed` / `failed` / *unchanged*	—	stays `generating` if any asset is still non-terminal
17	Admin preview	`GenerateStep.tsx`, `ReviewImportStep.tsx`	`GET …/ai-studio/:jobId` polled every 2 s while `analysing` / `generating`	—	`assetDto.imageUrl` = full-size S3 PNG	polling stops on any other status
18	Approve / select	`ReviewImportStep` → `PATCH …/assets/:assetId` (`updateAssetSelection`)	sync	`selected`, `is_featured`, `validation_accepted`	—	400 if selecting un-accepted failed/warning asset
19	Import	`ReviewImportStep.runImport` (sequential) → `POST …/assets/:assetId/import` → `importAsset`	sync, DB txn, `FOR UPDATE` locks	`product_images` ×9 rows; asset `imported=true`; job `importing`	PNG → 9 derivatives	per-asset idempotent; failures listed as "N failed", "Retry Failed Imports"

#	Step	File · function	Endpoint / job	Tables	Input → Output	Errors / retry
20	Derivatives	`imageProcessingService.processAndStoreImage`	inside import txn	S3 `products/<productId>/<uuid>-{thumbnail,small,medium,large}.{avif,webp}`, `-original.jpeg`	widths 150/400/800/1600, AVIF q85, WebP q75	S3 PUTs are not rolled back if the txn fails (orphans)
21	Finish	`completeImport`	`POST …/import/complete`	job `completed`	—	409 if a *selected* asset isn't imported
22	Product API	`productsService.getPublicProductBySlug` → `products.controller` DTO	`GET /api/v1/products/:slug`	`product_images ORDER BY sort_order`	all rows `{id,type,variant,format,url,isPrimary,sortOrder}`	—
23	Listing API	`PRIMARY_IMAGE_JOIN` (`products.repository.js`)	list/search/home endpoints	`is_primary AND variant='small' LIMIT 1`	`primaryImageUrl`	—
24	Storefront	`web/src/pages/PDPPage.tsx`, `features/pdp/ImageGallery.tsx`, `components/ProductCard.tsx`; SSR `web/server/index.js` + `pageCache.js`	SSR + client	`page_cache`	groups by sortOrder, AVIF→WebP, srcset 800w/1600w	SSR failure → client-only shell

5. Six-image logic (actual implementation)

Manifest function (`resolveAssetTypesForJob`, identical copy in `admin/src/features/aiStudio/generationRules.ts`):

Case	Manifest produced	Count	Spec expects
Non-Ring, Rose ON, presenter	YELLOW_FRONT, YELLOW_HERO_45, ROSE_FRONT, ROSE_HERO_45, PRESENTER_YELLOW_1, PRESENTER_ROSE	**6**	2Y cat + 2R cat + 1Y pres + 1R pres = 6 ✅
Non-Ring, Rose OFF, presenter	YELLOW_FRONT, YELLOW_HERO_45, PRESENTER_YELLOW_1, PRESENTER_YELLOW_2	**4**	4Y cat + 2Y pres = 6 ❌
Non-Ring, Rose ON, no presenter	2Y + 2R catalogue	**4**	6 ❌
Non-Ring, Rose OFF, no presenter	2Y catalogue	**2**	6 ❌
Necklace	Rose forced ON → as row 1 or 3	6 / 4	—
Ring, Rose ON	RING_HAND_1 (Y), RING_HAND_2 (**R**), RING_GOLD_FRONT, RING_GOLD_SIDE, RING_ROSE_FRONT, RING_ROSE_SIDE	**6**	✅ (2Y+2R cat, 1Y+1R "presenter" = hands)
Ring, Rose OFF	HAND_1, HAND_2 (Y), GOLD_FRONT, GOLD_SIDE	**4**	4Y cat + 2Y pres = 6 ❌

Count by layer:

Layer	Source	Value
Frontend expected	`generationRules.resolveAssetTypesForJob` ("Confirm & Generate N Images")	same as table — consistent with backend
Backend expected / manifest	`aiStudioService.resolveAssetTypesForJob`	same
Worker count	targets = all `PENDING` assets (or `assetIds` on retry)	= manifest
DB output records	`ai_studio_assets` (UNIQUE job+type)	= manifest. Production: 2,323 of 2,326 confirmed jobs have 6 rows; the other 3 have 4 (Rose ON is the localStorage default; only 1 confirmed job ever ran with Rose OFF)
Progress denominator	`GenerateStep`: `assets.length`; no % bar, per-tile status	= manifest
Import expected	`selected.length` (not the manifest)	**varies** — validation auto-deselects
Gallery order	browser import order: `display_order`, Ring hands +100 (last); appended after existing images (`MAX(sort_order)+1`)	not deterministic across retries/re-imports
Primary	`is_featured` asset → `is_primary`; default featured YELLOW_FRONT / RING_GOLD_FRONT	only if that asset is imported

Mismatches: (1) Rose OFF and no-presenter cases never produce six; (2) import count ≠ manifest; (3) gallery order is a client-side decision; (4) the `PRESENTER_YELLOW_2` slot only exists when Rose is OFF, so the same slot index means different shots depending on a flag; (5) `display_order` values overlap between Ring and non-Ring vocabularies (both start at 0).

6. Category rules (actual)

Common to every category: `FIDELITY_BLOCK` (preserve "...band or chain proportions, **clasps**..."), metal-colour sentence, satin catalogue background keyed by metal (`RING_GOLD_/RING_ROSE_CATALOGUE_BACKGROUND_NOTE`, now used for *all* categories), `CATALOGUE_NO_PERSON_NOTE` on non-Ring catalogue shots, generic validator. Output always 1024×1024 PNG regardless of prompt text.

Category	Catalogue views	Presenter / placement	Special rules	Hidden fastening	Validator extras	Auto-retry
Ring	True Front + ¾ Side (Gold, Rose)	2 **hand** shots, text-only identity (female default; Gents/Kids text variants)	`RING_FIDELITY_EXTRA`, hand placement negatives, hand background ≠ metal colour	n/a	per-shot orientation checklists (front/side/hand)	hands: 1 automatic retry on `failed`
Necklace	Front + 45° Hero (Y+R, forced)	neck & upper chest; library photo	set-earrings fidelity notes (catalogue vs presenter variants)	**none**	set-earring rules	no
Bracelet	Front + 45° Hero	one visible wrist	`BRACELET_CLASP_HIDDEN_NOTE`	**yes** (only category)	generic	no
Bangle	Front + 45°	one visible wrist	—	none	generic	no
Pendant	Front + 45°	neck	—	none	generic	no
Earrings	Front + 45°	ears	—	n/a	generic	no
Chain	Front + 45°	**generic** "its natural position"	—	**none**	generic	no
Anklet	Front + 45°	**generic** (template says "ankle-focused")	—	**none**	generic	no
Mangalsutra	Front + 45°	**generic**; excluded list forbids "necklaces, pendants"	—	none	generic — systematically reports "Expected Mangalsutra, detected Necklace/Pendant"	no
Nose pin / Brooch / Other	Front + 45°	generic	—	none	generic	no

Templates (`ai_studio_category_templates`, all active): every category has the **same** `front_prompt` — "*...on a seamless neutral studio background... no props, no people*" — and `presenter_crop = 'portrait 4:5'`.

Hidden-fastening check: Necklace ❌, Bracelet ✅, Anklet ❌. Bracelet's rule also directly contradicts `FIDELITY_BLOCK` ("preserve ... clasps").

Crop/position: catalogue prompts say 816×816 (`CATALOGUE_SIZE_NOTE`, `HERO_45_PROMPT`, `RING_SIZE_NOTE`); presenter says 4:5; the API is always called with `size:'1024x1024'`.

7. Customer Category and presenter logic

Customer category	Non-Ring presenter shots	Ring hand shots	Identity reference
Women (default)	presenter library photo attached as 2nd image (`front_portrait_url` for Y1/Rose, `face_45_url` for Y2) + its `prompt_descriptor`	hardcoded female text	photo for presenter shots; **text-only** for hands
Unisex	identical to Women (no branch exists)	identical to Women	same
Gents	library photo **not** attached; one master portrait per job via `images.generate`, stored in `gents_master_presenter_key`, attached as image #1	Gents hand text	photo for presenter shots; **text-only** for hands
Kids	library photo not attached; text descriptor only	Kids hand text	**text-only** everywhere

Does customer category leak into other dimensions?

Dimension	Affected?
Image count / catalogue views / metal distribution / order / progress / gallery positions	No (manifest ignores it) ✅
Catalogue **prompts**	**Yes** — `locked.confirmedCustomerCategory` ("The confirmed customer category is Kids/Gents.") is added to every asset, including product-only catalogue shots. Possibly related: 8 of the 10 Kids safety rejections were catalogue shots.
Validation	Adds demographic checks to presenter/hand shots only ✅

Consistency: Ring `RING_HAND_1` and `RING_HAND_2` are independent text-only generations — two different people/hands are expected. Kids presenter shots likewise. Gents identity mismatch was flagged 73 times; 64 were accepted and imported anyway.

8. Prompt inventory and conflicts

Source	File · constant	Applies to	Order in final prompt
Product identity / category / customer category	`buildAssetPromptSections` `locked.*`	all	1–3
Global fidelity	`FIDELITY_BLOCK`	all	4
Metal colour	`metalColourNote`	all	5
Composition	`CATALOGUE_SIZE_NOTE` + template `front_prompt`; `HERO_45_PROMPT`; Ring front/side literals; presenter `template.presenter_placement/crop`; `RING_SIZE_NOTE` + `RING_HAND_PLACEMENT_COMPOSITION`	per asset type	6
Presenter pose	presenter `prompt_descriptor`, `GENTS_/KIDS_PRESENTER_DESCRIPTOR`	presenter shots	8
Category placement	`PLACEMENT_RULES / GENERIC_PLACEMENT_RULE`; Ring hand texts; `GENTS_IDENTITY_LOCK_NOTE`	presenter/hand	9
Background	`RING_GOLD/ROSE_CATALOGUE_BACKGROUND_NOTE`, `PRESENTER_BACKGROUND_NOTE`, `RING_HAND_BACKGROUND_NOTE`, Gents/Kids variants	all	10
Lighting	`RING_CATALOGUE_LIGHTING_NOTE`, `RING_HAND_LIGHTING_NOTE`	catalogue/hand	11
Admin custom	`custom_creative_instructions` (6 fields, ≤500 chars each)	any	replaces creative fields
Negatives	per-type list + `RING_FIDELITY_EXTRA`, `BRACELET_CLASP_HIDDEN_NOTE`, `NECKLACE_SET_NOTE_*`, `CATALOGUE_NO_PERSON_NOTE`, audience negatives	per type/category	13
Regeneration	`priorFailure` → "A previous attempt was rejected: … Correct this specific issue." (prompt rebuilt from code, **not** from the persisted prompt)	retries after warning/failed	appended to negatives
Output spec	`locked.outputSpecs`	all	last
Quality check	`VALIDATION_SYSTEM_PROMPT` + `ring/necklace/catalogue/audienceValidationContextFor`	all	separate call
Analysis	`ANALYSIS_SYSTEM_PROMPT`	analyse	separate call
Gents identity	`GENTS_MASTER_PRESENTER_PROMPT`	Gents	separate call
Image Enhancement	`imageEnhancementService.js` `ENHANCEMENT_PROMPT`	manual enhance tool	separate feature
Product Size	`productSizeImageService.js` `PRODUCT_SIZE_BASE_PROMPT` + its own validator	size image	separate pipeline

There's no prompt versioning: `prompt_version` / `template_version` are hard-coded `'v1'` while the prompt text has changed many times.

Conflicts found (confirmed in the assembled prompts stored in production)

1. **Background, three-way contradiction (every non-Ring catalogue shot):** template says *"seamless neutral studio background… no props"*, then *"soft champagne-ivory satin fabric backdrop with visible folds"*, then the negative *"Do not add props, text, hands, packaging, flowers, fabric or decorative elements."* The 45° Hero adds a second "…**fabric**…" exclusion.
2. **Clasp:** `FIDELITY_BLOCK` "preserve … clasps" vs `BRACELET_CLASP_HIDDEN_NOTE` "do not show the clasp".
3. **Size/aspect:** "816 × 816px" and "portrait 4:5 crop" vs actual `1024x1024`.
4. **Mangalsutra presenter:** *"Place only this exact mangalsutra on the selected presenter's its natural position. No other jewellery of any kind — no … necklaces, pendants …"* — a grammatically broken placement, plus an exclusion list that forbids the product's own form (a mangalsutra *is* a necklace with a pendant). The validator then fails 195 of 390 Mangalsutra images on category.
5. **Presenter lighting:** the presenter `prompt_descriptor` contains "warm ambient lighting" while the background note asks for a solid studio backdrop.
6. **Duplication:** "Photorealistic, premium e-commerce… No text, logos, or watermarks." appears twice in every prompt; "Do not redesign, reinterpret…" appears twice; the excluded-jewellery sentence appears twice in presenter prompts.
7. **Necklace/Anklet:** no hidden-fastening rule despite the stated requirement.

Sanitized example — catalogue (YELLOW_FRONT, Mangalsutra, stored prompt)

```
Use the uploaded product reference as the only jewellery design. The confirmed product category is Mangalsutra.
The uploaded jewellery reference is the definitive product. Preserve its exact jewellery type, … clasps, … Photorealistic,
premium e-commerce jewellery photography. No text, logos, or watermarks. The metal colour is yellow gold — preserve it exactly.
Create a premium square jewellery catalogue photograph at 816 × 816px. A clean, straight-on catalogue product photo on a
seamless neutral studio background, evenly lit, centred, no props, no people. Use a soft champagne-ivory satin fabric backdrop
with visible natural folds and drape — base colour #F4E7DA … Use soft, premium studio lighting … Do not redesign … This is a
product-only catalogue image: do not include any person … Do not add props, text, hands, packaging, flowers, fabric or
decorative elements. Photorealistic, premium e-commerce jewellery photography. No text, logos, or watermarks.
```

Sanitized example — presenter (PRESENTER_YELLOW_1, Mangalsutra)

```
… The metal colour is yellow gold — preserve it exactly. Framed as a traditional neck portrait presenter shot (portrait 4:5 crop).
Use realistic product scale and placement. Worn naturally by a traditional presenter with graceful ethnic/festive styling and warm
ambient lighting, matching the attached presenter reference photo. Place only this exact mangalsutra on the selected presenter's its
natural position. No other jewellery of any kind — no rings, earrings, necklaces, pendants, … Use a soft, solid studio backdrop colour
that elegantly complements the presenter … Do not replace the uploaded product with a similar design. Photorealistic, …
```

Sanitized example — regeneration (retry of a failed RING_GOLD_FRONT)

When an asset with `validation_status` ≠ passed is retried, `promptOverride` is skipped and the prompt is **rebuilt from current code** (`buildPrompt`). If the code changed since the job was confirmed, the retry silently uses the new text:

```
<same locked + composition + background sections as the original> … Do not use the same orientation as the Three-Quarter Side View
shot … A previous attempt was rejected: Failed: Image is not a true front view; the ring is at a three-quarter angle.
Correct this specific issue. Photorealistic, premium e-commerce jewellery photography. No text, logos, or watermarks.
```

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9. Storage architecture

Item	Value
Reference images	`products/ai-studio/<productId>/references/<uuid>.jpeg` (re-encoded JPEG q92)
Raw generated	`products/ai-studio/<jobId>/<uuid>.png` (1024×1024, ~1–2 MB) — slot is **not** in the key; only `ai_studio_assets.image_key` links it
Gents identity	`products/ai-studio/<jobId>/gents-master-<uuid>.png`
Final derivatives	`products/<productId>/<uuid>-{thumbnail,small,medium,large}.{avif,webp}` + `-original.jpeg` (widths 150/400/800/1600, `withoutEnlargement` → a 1024 source yields a 1024-wide "large")
MIME	from extension in `storageProvider.s3.save`
Access	public-read bucket policy; no signed URLs anywhere
Cache headers	new uploads `public, max-age=31536000, immutable`; most pre-existing objects have none (separate pending S3 task)
DB URL = API URL = rendered URL	yes — `https://boxgold.s3.ap-south-1.amazonaws.com/<key>`, unchanged end to end
Temp files	none on disk (all buffers in memory)

Verification (read-only): - `HEAD` of every `product_images` large+small AVIF URL (15,798 objects): **15,798 × HTTP 200** `image/avif`. - The 400 most recent `ai_studio_assets.image_key` objects: **400 × 200** `image/png`. - `product_images`: 0 non-S3 URLs, 0 signed URLs, 0 local URLs; 7,899 groups all have exactly 9 rows; 0 duplicate variant/format rows; 0 groups with mixed `is_primary`; 0 products with >1 primary; 0 imported assets without `image_key`.

Conclusion: storage, URLs, MIME types and derivatives are healthy. Images are not disappearing at the storage layer.

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10. Database relationships

```
products 1—* product_images (type ORIGINAL|AI_GENERATED|PRODUCT_SIZE, variant, format, url, is_primary, sort_order)
products 1—* ai_studio_jobs (status, reference_image_urls jsonb, analysis, jewellery_type, customer_category,
                             generate_rose_gold, presenter_id→presenters, gents_master_presenter_key)
ai_studio_jobs 1—* ai_studio_assets (asset_type, display_order, status, image_key, assembled_final_prompt,
                             validation_status/result/accepted, selected, is_featured, imported, retry_count, error)
ai_studio_category_templates (jewellery_type → front_prompt, presenter_placement, presenter_crop, hero_45_prompt)
legacy: ai_image_jobs 1—* ai_generated_images ; ai_studio_presenters → renamed `presenters`
```

There is **no link** from a `product_images` row back to the `ai_studio_assets` row it came from, and no metal-colour or slot column on `product_images`. After import, the system can't tell which gallery image is "Rose Gold Front".

Production counts:

Check	Result
Jobs	completed 2,006 · cancelled 311 · review_ready 66 · failed 42 · partially_failed 5 · importing 1
READY assets **never imported**	completed jobs **1,092** (675 still need "Accept"), review_ready 396, partially_failed 17, cancelled 1,292
Completed jobs whose featured asset was not imported	**180**
Latest completed job per product, imported < 6	137 products (59 imported 2/6, 40 imported 1/6)
Assets stuck `GENERATING`	86 (all in cancelled jobs) · `PENDING` 32 (cancelled/failed jobs)
Job stuck `importing` since 2 Sep (blocks new jobs for that product)	1
Validation outcome on READY assets	Ring: **3,432 failed / 1,588 passed / 124 warning** · Mangalsutra 195 failed / 165 passed
Failed-validation assets accepted & imported anyway	3,394 (+2,524 warnings)
Products with AI images where PDP first image ≠ featured/primary	**906 of 1,287** published
Products with >1 completed job	580 (481 with two jobs)
Products where imported AI images were later removed	510
Shared image URLs across products (attach-existing)	9 URLs / 18 rows
AI products showing an old non-primary ORIGINAL first on PDP	2
Primary=ORIGINAL while AI images exist	1 (DRAFT)
Published products with no images	0

11. Display pipeline investigation

Example A — successful: Giardinonia Diamond Ring (job `00d14adb…`, Ring, Women, Rose ON)

Stage	Result
Manifest	6 Ring assets ✅
Generation	all 6 READY, 33–79 s each, retry_count 0
Validation	**5 of 6 failed** (e.g. "Not a true front view", "Expected Yellow, detected Rose"), all "Accepted Anyway"
Import	6/6 imported 18:25–18:26 (~13 s per asset), sort 0–5, primary = sort 2
Storage	all objects 200
API	6 groups, 9 rows each ✅
Storefront	PDP opens on sort 0 (Yellow Front); listing card shows primary (the admin-featured **Rose** Front). The product's "first image" differs between the listing and the PDP.

Example B — problematic: Rovella Diamond Tanmaniya with Chain (job `f15167d6…`, Mangalsutra, 19 Oct)

Stage	Result
Manifest	6 assets ✅
Generation	all 6 READY
Validation	**6 of 6 failed** — "Expected Mangalsutra, but Necklace/Pendant was detected" (false failure)
Selection	worker set `selected=false` on all six; admin accepted only YELLOW_FRONT and ROSE_FRONT
Import	**2 of 6 imported**; job `completed`; 4 generated images never reached the product
Product gallery	6 groups = 2 new images (sort 0, 2, created 18:12) + 4 images from an **earlier run** (sort 1, 4, 5, 6, created 07:37–07:41); sort 3 missing
API / storefront	serves this mixed set exactly as stored; images display, but they come from two runs with different backgrounds and poses

Where the image stops: at **Review & Import selection** (N1). The trigger is a validation false failure, followed by auto-deselection. Classification: *Validation failure → Approval/import failure*, then *Gallery-order failure*.

Failure classification

Failure class	Status	Evidence
Approval/import failure (generated, never imported)	**Confirmed**	1,092 + 396 + 17 READY unimported assets; 180 completed jobs without featured import
Validation failure (false negatives drive N1)	**Confirmed**	Ring 66 % fail rate; Mangalsutra category false-fails; 3,394 failures overridden by admins
Gallery-order failure (PDP opens on non-featured image; mixed runs; holes)	**Confirmed**	906/1,287 products; Example B
Generation failure (OpenAI credits exhausted, shown as "busy")	**Confirmed**	431 log lines "429 You have no credits remaining"; 85 job + 79 asset errors labelled "busy"
Generation failure (safety rejection shown as "image could not be processed")	**Confirmed**	46 log lines "rejected by the safety system"; 10 Kids assets
CDN/cache failure (stale page after import)	**Highly likely** (≤10 min)	no `invalidateProductPages` call in any image mutation; `page_cache` TTL 600 s
Progress failure (assets stuck GENERATING)	**Confirmed** (cancelled jobs only) / **Possible** (worker restart)	86 assets
SSR rendering failure ("Too Many Requests" → client-only shell)	**Confirmed historical, resolved**	1,615 entries in `boxdiamonds-web-error.log`, last written 6 Oct; loopback is now exempt in `middleware/rateLimit.js`
Storage / URL / API serialization failure	**Not found**	§9 checks all pass
Product-variation selection failure (rose gold selected but gallery doesn't switch)	**Confirmed by design**	no metal/slot data on `product_images`; gallery never filters by gold colour

12. Root-cause findings

Critical — images generated but never shown

C-1 Validation auto-deselection silently drops generated images from import — *Confirmed* - Evidence: `aiStudioJob.js generateOneAsset` sets `selected:false` for any non-passed result; `ReviewImportStep.handleImport` imports only `selected`; `completeImport` closes the job once every *selected* asset is in. Data: 1,092 READY assets unimported in completed jobs, 180 completed jobs without the featured image, Example B. - User effect: the admin sees "Import 2 Images" and believes the job is done; 4 of 6 images never appear on the product. - Root cause: an unreliable validator is wired as a gate, and the gate's effect (deselection) isn't surfaced as a blocking decision. - Fix: never complete a job while READY assets are undecided. Each slot must be explicitly *Imported* or *Discarded*. Show "4 images need a decision" as a blocking state, not a hidden checkbox. - Risk: low. Complexity: S–M.

C-2 Validator false failures (Ring orientation, Mangalsutra category) — *Confirmed* - Evidence: Ring 3,432/5,170 failed; 195/390 Mangalsutra failed, mostly "Expected Mangalsutra, detected Necklace/Pendant"; 3,394 failures overridden by admins. - User effect: feeds C-1, and validation has become noise that admins click through. - Root cause: the Mangalsutra category is visually a necklace with a pendant, so the enum-level comparison can't work; the Ring orientation checklist is stricter than the generator's real output; one vision call judges its own image without sibling context. - Fix: per-category equivalence (Mangalsutra ⊃ Necklace/Pendant); treat orientation issues as `warning`; report a confidence per check; measure precision against admin decisions before using validation as a gate. - Risk: low. Complexity: M.

C-3 PDP gallery opens on a different image than the featured/listing image — *Confirmed* - Evidence: `ImageGallery.groupPhotos` sorts by `sortOrder` and starts at index 0; `is_primary` is used only by the listing join. 906 of 1,287 published AI products differ. - User effect: customers click a Rose Gold card and land on a Yellow Gold photo (Giardinonia), or see a hand shot or old photo first. - Root cause: two independent notions of "first image" (`is_primary` and `sort_order`), and ordering decided by browser import order. - Fix: make import put the featured asset at the lowest `sort_order` (or make the gallery start on the primary), and assign gallery position server-side from the slot. - Risk: low. Complexity: S.

High

H-1 No cache invalidation for any image mutation — *Confirmed (code); Highly likely impact* - Evidence: `invalidateProductPages` is never called from `importAsset`, `completeImport`, `uploadOriginal`, `attachExisting`, `setPrimary`, `reorder`, `remove` or `approve`. - User effect: for up to 10 min after import, SSR HTML (PDP, listing, home) shows old or no images; the client refetch usually corrects the PDP after hydration, while crawlers and first paint see stale content. - Fix: call `safeInvalidateProductPages(product)` after each of these mutations (and once in `completeImport`). - Risk: low. Complexity: S.

H-2 Partial imports mix generations; gallery appends and leaves holes — *Confirmed* - Evidence: `MAX(sort_order)+1` appends; 580 products had more than one completed job; Example B mixes two runs and has a missing position. - Fix: slot-based import that *replaces* the slot's previous image (keeping the old one for rollback), with a deterministic slot→position map. - Complexity: M.

H-3 Rose OFF / no-presenter manifests are not six images — *Confirmed (code)* - Evidence: see §5. - Fix: one authoritative 6-slot manifest per (category, rose flag). - Complexity: M (prompts for 2 extra Yellow views).

H-4 Prompt contradictions (background/fabric, clasp, size, Mangalsutra placement) — *Confirmed* - Evidence: §8 and the stored prompts. - User effect: inconsistent backgrounds (flat vs satin), visible clasps, mis-placed Mangalsutra, and validator confusion. - Fix: a rule builder with explicit precedence; remove the template background text; add hidden-fastening rules for Necklace, Anklet and Chain/Mangalsutra if wanted; fix the generic placement grammar and the excluded list. - Complexity: M.

H-5 Delete removes S3 objects that other products still reference — *Confirmed (code), latent* - Evidence: `productImages.controller.remove` deletes `storageProvider.delete(keyFromUrl(url))` for every row without checking other `product_images` rows. `attachExisting` shares URLs (9 URLs / 18 rows exist today). - User effect: deleting a shared photo from product A breaks it on product B. - Fix: delete the object only when no other row references the URL. - Complexity: S.

H-6 Primary rules differ between paths — *Confirmed* - Evidence: `uploadOriginal` treats "first ORIGINAL" as the trigger, so a manual upload to an AI-only product steals primary. The legacy `approve` path uses "no images at all". AI Studio uses "featured imported". `PDPPage` hides only `ORIGINAL && isPrimary`, so after the AI featured import an old manual photo becomes visible again (2 products). - Fix: a single primary policy in one repository function. - Complexity: S.

Medium

M-1 OpenAI billing exhaustion reported as "busy, try again" — *Confirmed* - Evidence: `mapOpenAiError` maps every `RateLimitError` (including `insufficient_quota`) to "busy"; 431 log entries. - Fix: distinguish quota from rate limit, alert, and stop the queue. - Complexity: S.

M-2 Safety rejections reported as "uploaded image could not be processed" — *Confirmed* - Evidence: 46 log entries; 10 Kids assets, 8 of them catalogue shots whose prompt contains "customer category is Kids". - Fix: map `content_policy` separately, and drop the customer-category sentence from catalogue prompts. - Complexity: S.

M-3 No reaper for non-terminal assets/jobs — *Confirmed (86 GENERATING in cancelled jobs); Possible after restarts* - Evidence: a pm2 restart or pg-boss expiry mid-run leaves assets `GENERATING`, which first-run retries skip (they target only `PENDING`). The job then never leaves `generating`. Worst-case Ring-hand time (2 attempts × 600 s × SDK retry) exceeds the 30-min expiry → *Possible* duplicate run. - Fix: heartbeat/timeout reaper, idempotent per-slot generation, expiry sized to real worst case. - Complexity: M.

M-4 Job stuck in `importing` blocks new jobs — *Confirmed (1 job)* - Evidence: `findActiveJobByProduct` includes `importing`. - Fix: auto-complete when all selected assets are imported. - Complexity: S.

M-5 Gallery order is a client decision — *Confirmed* - Evidence: `importSortKey` lives in `ReviewImportStep.tsx`; "Retry Failed Imports" appends retried slots after the others. - Complexity: S (move to server).

M-6 Regeneration rebuilds the prompt from current code, not the confirmed prompt — *Confirmed* - Evidence: `generateOneAsset` sets `promptOverride` undefined whenever `priorFailure` is set. - Complexity: S.

M-7 Validation call failures are swallowed (status `null`) — *Confirmed* - Evidence: 61 READY assets. - Effect: the image passes the import gate unvalidated.

Low

- **L-1** Manifest, metal mapping and catalogue type sets are duplicated in 3–4 places: `aiStudioService.js`, `aiStudio.controller.js`, `generationRules.ts`, `ReviewImportStep.tsx`, plus the DB CHECK.
- **L-2** `prompt_version` / `template_version` are hard-coded `'v1'`, so there's no traceability.
- **L-3** The legacy `ai-images` stub pipeline is still routable.
- **L-4** The listing `PRIMARY_IMAGE_JOIN` picks AVIF or WebP arbitrarily (`LIMIT 1`, no `ORDER BY`; 1,292 products have both).
- **L-5** The admin preview loads full 1–2 MB PNGs for every tile.
- **L-6** Three image-object shapes: admin grouped `{sortOrder, variants[]}`, storefront raw rows, studio `imageUrl`.
- **L-7** S3 PUTs happen inside the DB transaction, so failed imports leave orphan objects.
- **L-8** Comments describe removed features (e.g. "Hand Pose" choice, "no upload flow exists yet").

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13. Target architecture

Component	Responsibility	Input → Output	Failure	Retry
Reference Analyzer	type/metal/stones/customer suggestion	reference → analysis + confidences	job `needs_input`, explicit error code	2× backoff on 429/5xx; none on quota
Category Resolver	final jewellery type + equivalence groups (Mangalsutra≈Necklace/Pendant)	analysis + product category + admin choice → `jewelleryType`	blocks until confirmed	—

Component	Responsibility	Input → Output	Failure	Retry
Customer Category Resolver	presenter identity policy only	admin choice → `{presenterSource, identityRef}`	—	—
Output Manifest Builder	**one** server-owned 6-slot manifest `(slotId, view, metal, kind, galleryPosition)`	type + rose flag → 6 slots	invalid combo → 400	—
Prompt Rule Builder	layered rules with precedence (fidelity > category > view > metal > background > creative), conflict lint, versioned	slot + rules version → prompt + `prompt_version` hash	lint failure blocks confirm	—
Generation Orchestrator	dependent ordering (identity → presenter shots), per-slot state machine, heartbeats	manifest → slot jobs	slot `failed` with code	per-slot, bounded, idempotent by `(job, slot, attempt)`
AI Provider Adapter	OpenAI calls, error taxonomy (quota / rate / safety / bad-input / 5xx)	request → buffer or typed error	quota halts the queue and alerts	rate/5xx only
Identity Reference Manager	one identity per job per audience (incl. Women ring hands, Kids)	category → stored identity image	—	regenerate only on explicit admin action
Quality Validator	advisory scores per check, category-aware, calibrated	image + ref → checks[]	errors recorded, never null-silent	1×
Asset Processor	derivatives (AVIF/WebP sizes + original) **before** the DB txn	PNG → derivative set	cleanup on failure	1×
Storage Adapter	write-once keys incl. slot `…/<slot>/<uuid>`, ref-counted delete	buffer → key/url	—	SDK
Product Image Importer	slot-aware replace (old kept as `archived`), server-assigned position, single primary policy, `source_asset_id` link	slot decisions → product_images	txn rollback + orphan cleanup	idempotent
Cache Invalidator	product, category, home `page_cache` after any image mutation	productId → DELETEs	logged, non-fatal	—
Audit Logger	who/what/when per slot: prompt version, model, validation, decision	events → append-only table	—	—

14. Prioritized roadmap

Phase	Changes	Dependencies	Risks	Tests	Rollback	Expected result
1 Display reliability	(a) block job completion while READY slots are undecided; show unimported count; (b) import featured slot at position 0 / gallery starts at primary; (c) `safeInvalidateProductPages` on all image mutations; (d) ref-counted S3 delete; (e) auto-complete stuck `importing`; (f) quota/safety error messages	none	low; UI flow change for admins	unit: import ordering, invalidation called; e2e: generate → import → PDP/listing show same first image	revert commits; no schema change	every generated image is either visibly imported or explicitly discarded; listing = PDP first image; no 10-min stale pages
2 Manifest & ordering	server-owned 6-slot manifest for all flags; slot column + `source_asset_id` on `product_images`; slot-aware replace; remove client `importSortKey`	Phase 1	migration on 70k rows (additive, nullable)	manifest matrix tests per category × rose × presenter	columns nullable; old path kept behind a flag	deterministic 6 images in fixed gallery positions
3 Product fidelity	rule builder with precedence + lint; remove template background text; hidden-fastening for Necklace/Anklet/Chain; Mangalsutra placement; drop customer-category line from catalogue prompts; real size/aspect	Phase 2	visual regressions	golden-prompt snapshot tests; 20-product visual review	prompt version pin	consistent backgrounds, correct fastening, fewer re-rolls
4 Presenter consistency	identity references for Women ring hands, Kids, Unisex; sequential dependent generation	Phase 3	safety filter on Kids	identity-match validator precision	per-audience flag	same person across presenter/hand shots
5 Validation & retry	category-aware, calibrated, advisory validator; slot-specific regeneration using persisted prompt + correction; reaper/heartbeats	Phases 2–3	over-tight gates again	measure against 3,394 historical overrides	validator → advisory only	validation that admins trust; no stuck jobs
6 Performance & cost	derivatives outside txn, parallel S3 PUTs; thumbnail previews in admin; stop on quota; dedupe re-runs; retire legacy stub pipeline; clean orphaned/cancelled assets (after approval)	any	deleting data	storage diff dry-run	snapshots	faster import (~13 s → ~3 s/asset), lower OpenAI spend

15. Open questions

1. Should Rose OFF really produce **4 Yellow catalogue views**? Today there are only two Yellow catalogue prompts (Front, 45°/Side). Which two new views are wanted?
2. For non-Ring products with **no presenter**, should the 2 presenter slots be replaced (e.g. by extra catalogue views) or stay absent?
3. Should the gallery open on the **featured image**, or on a fixed slot order (e.g. Yellow Front first) with the listing card following it?
4. Should re-generation **replace** a product's previous AI images per slot, or keep appending as today?

5. Is Mangalsutra intended to show a visible chain clasp? And Necklace/Anklet clasps: hide or show?
6. How should the 1,092 unimported READY images in completed jobs, and 66 abandoned review jobs, be handled: reviewed, imported, or discarded? Nothing has been touched.
7. Should validation remain a gate at all, given admins accepted ~72 % of failed images (3,464 of 4,797)?
8. Is the OpenAI account on auto-recharge? The logs show repeated "no credits remaining" periods.